

Aligned-Center Criteria for Products with Multiple q -Integer Spacers

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Abstract

Connelly, Ito, Martinez, Shevchenko, and Yang conjectured a sufficient condition for the unimodality of a product of ordinary q -integers and one lacunary factor. We first prove that condition for every number of ordinary factors and every spacer step in its stated range. We then prove a hybrid criterion for arbitrarily many lacunary factors: disjoint divisibility pairs are absorbed factor by factor, and the remaining spacer increments are assigned to the remaining ordinary factors by a nonnegative integer allocation matrix. No coprimality among the spacer steps is required. Both results follow from one aligned-center recursion. The criterion is sufficient rather than necessary, and explicit small examples map that boundary.

1 Introduction

For a positive integer n , write

$$[n]_q = 1 + q + \cdots + q^{n-1}.$$

Connelly, Ito, Martinez, Shevchenko, and Yang proposed the following criterion [1, Conjecture 5.4]. If $r \geq 2$, $k \geq 1$, and $a_1, \dots, a_k, b$ are positive integers satisfying

$$r \mid a_i \quad \text{for some } i, \quad \text{or} \quad b \leq 1 + \sum_{i=1}^k \left\lfloor \frac{a_i}{r} \right\rfloor, \quad (1)$$

then

$$\prod_{i=1}^k [a_i]_q [b]_{q^r} \quad (2)$$

is unimodal. They conjectured necessity when $k \leq 3$ or $r \leq 3$, having already proved the full equivalence for $k = 1$ and for $(k, r) = (2, 2)$. Their bounded checks are separate computational evidence.

Our first result settles the sufficient direction exactly as stated.

Theorem 1 (one-spacer criterion). *Let $r \geq 2$, $k \geq 1$, and let $a_1, \dots, a_k, b$ be positive integers. If either condition in (1) holds, then the polynomial (2) is symmetric and unimodal.*

The same mechanism permits several spacers. For the next statement, choose sets $J \subseteq \{1, \dots, s\}$ and $I \subseteq \{1, \dots, k\}$, and a bijection $\phi : J \rightarrow I$. Write J^c and I^c for their complements.

Theorem 2 (hybrid multi-spacer criterion). *Let $k, s \geq 1$; let $a_1, \dots, a_k, b_1, \dots, b_s$ be positive integers; and let $r_1, \dots, r_s \geq 1$. Suppose that*

$$r_j \mid a_{\phi(j)} \quad (j \in J),$$

and that there are nonnegative integers $d_{j,i}$, for $j \in J^c$ and $i \in I^c$, such that

$$\sum_{i \in I^c} d_{j,i} = b_j - 1 \quad (j \in J^c), \quad (3)$$

and

$$a_i - \sum_{j \in J^c} r_j d_{j,i} \geq 1 \quad (i \in I^c). \quad (4)$$

Then

$$\prod_{i=1}^k [a_i]_q \prod_{j=1}^s [b_j]_{q^{r_j}}$$

is symmetric and unimodal. Empty residual index sets are allowed, with the displayed conditions interpreted as empty sums.

Theorem 2 has no coprimality hypothesis. It is genuinely beyond the named conjecture because it allows arbitrarily many spacers, but it remains only sufficient. The source example $([3]_q)^4 [2]_{q^4}$ is unimodal while violating (1), so no general necessity claim is made. At two spacers, $[2]_{q^2} [2]_{q^3} = 1 + q^2 + q^3 + q^5$ is symmetric but not unimodal and is excluded by (4). The nearby unimodal product $[5]_q [2]_{q^2} [2]_{q^3}$ is also outside (4), showing that the new condition is not necessary. We do not give an exact characterization of multi-spacer unimodality.

2 Symmetric unimodality

A finite nonnegative coefficient sequence is *symmetric unimodal about C* if coefficients at exponents equidistant from C agree and the coefficients weakly increase up to C . We allow initial and terminal zeros; this convention makes translates convenient.

Lemma 3. *If F and G are nonnegative symmetric unimodal polynomials about centers C_1 and C_2 , respectively, then FG is symmetric unimodal about $C_1 + C_2$. A sum of nonnegative symmetric unimodal polynomials having a common center C is symmetric unimodal about C .*

Proof. The assertion about sums follows coefficientwise. For products, decompose each symmetric unimodal coefficient sequence into a nonnegative linear combination of indicator sequences of nested intervals having the same center. Unimodality permits zeros only before and after the nonzero support, not internally, which licenses this nested-interval decomposition. The convolution of two interval indicators is a symmetric trapezoidal sequence, hence unimodal, and all resulting convolutions have the sum of the original centers. Summing them proves the claim. $\square$

3 The aligned-center recursion

Lemma 4 (aligned-center recursion). *For positive integers a, b, r ,*

$$[a + r]_q [b + 1]_{q^r} = q^r [a]_q [b]_{q^r} + [a + r(b + 1)]_q. \quad (5)$$

Proof. Use

$$[a + r]_q = [r]_q + q^r[a]_q, \quad [b + 1]_{q^r} = 1 + q^r[b]_{q^r}.$$

After subtracting the first term on the right side of (5), the remaining terms are

$$\begin{aligned} [r]_q[b + 1]_{q^r} + q^r[a]_q(1 + (q^r - 1)[b]_{q^r}) &= [r(b + 1)]_q + q^{r(b+1)}[a]_q \\ &= [a + r(b + 1)]_q. \end{aligned}$$

For a coefficient-level proof independent of this manipulation, the left side counts pairs (x, j) with $0 \leq x < a + r$ and $0 \leq j \leq b$, weighted by $x + rj$. The pairs with $x < a$ and $j \geq 1$ give $q^r[a]_q[b]_{q^r}$. The complementary pairs have either $j = 0$, or $x \geq a$ and $j \geq 1$; their weights occur once each and fill the consecutive interval from 0 to $a + r(b + 1) - 1$. They give the second term in (5). $\square$

4 The hybrid criterion

We first prove the allocation part of Theorem 2.

Lemma 5 (matrix induction). *Let $k \geq 1$ and $s \geq 0$. If nonnegative integers $d_{j,i}$ satisfy (3)–(4) with $I = J = \emptyset$, then*

$$\prod_{i=1}^k [a_i]_q \prod_{j=1}^s [b_j]_{q^{r_j}}$$

is symmetric and unimodal.

Proof. Induct on s . The case $s = 0$ is a product of ordinary q -integers and follows from Lemma 3. For $s \geq 1$, put

$$a_i^{(0)} = a_i - r_1 d_{1,i}.$$

Condition (4) shows that the rows $2, \dots, s$ certify the induction hypothesis for

$$\prod_i [a_i^{(0)}]_q \prod_{j=2}^s [b_j]_{q^{r_j}}.$$

This is the base at first-spacer length one.

Reconstruct the $b_1 - 1$ increments in any order realizing the first row of the matrix. At one step, let x be the current selected ordinary length, c the current first-spacer length, $B(q)$ the product of the other ordinary factors, and $R(q)$ the product of spacers $2, \dots, s$. Lemma 4 gives

$$\begin{aligned} [x + r_1]_q B(q) [c + 1]_{q^{r_1}} R(q) \\ = q^{r_1} [x]_q B(q) [c]_{q^{r_1}} R(q) + [x + r_1(c + 1)]_q B(q) R(q). \end{aligned} \tag{6}$$

The first summand is a translate of the preceding polynomial. In the second, every ordinary length is at least its base value, and the selected length is larger still. The unchanged rows $2, \dots, s$ therefore continue to satisfy (4), so the induction hypothesis makes the correction term symmetric unimodal.

If the old degree is

$$E = (x - 1) + \deg B + r_1(c - 1) + \deg R,$$

the new degree is $E + 2r_1$. The translated old support has endpoints r_1 and $E + r_1$, while the correction term has degree

$$x + r_1(c + 1) - 1 + \deg B + \deg R = E + 2r_1.$$

Both summands in (6) therefore have center $(E + 2r_1)/2$. Their sum is symmetric unimodal by Lemma 3. Iteration reconstructs the first spacer and all ordinary lengths, completing the induction. $\square$

Proof of Theorem 2. For every $j \in J$, write $a_{\phi(j)} = r_j e_j$ and set $z = q^{r_j}$. Then

$$[a_{\phi(j)}]_q [b_j]_{q^{r_j}} = [r_j]_q ([e_j]_z [b_j]_z) \Big|_{z=q^{r_j}}. \quad (7)$$

The product $[e_j]_z [b_j]_z$ is symmetric unimodal in z by Lemma 3. Substitution followed by multiplication by $[r_j]_q$ repeats each coefficient in a consecutive block of length r_j , so (7) is symmetric unimodal in q .

The map ϕ is a bijection between the selected sets, so each ordinary factor is consumed by at most one spacer. Thus the factors in (7) are disjoint and do not interact with the residual allocation matrix. The residual product is symmetric unimodal by Lemma 5; multiplying it by the pair factors preserves symmetric unimodality by Lemma 3.

If no residual ordinary factors remain, (3) forces every residual spacer length to equal one, so that residual product is the constant polynomial. This also covers the empty case. $\square$

The statement permits $r_j = 1$. Such a spacer is an ordinary bracket. In the one-factor, one-spacer matrix case, (3)–(4) become $b \leq a$; swapping the two ordinary brackets covers the opposite ordering. No step of the proof assumes $r_j \geq 2$.

5 The named conjecture as a corollary

Proof of Theorem 1. If $r \mid a_i$ for some i , pair that ordinary factor with the sole spacer in Theorem 2.

Otherwise, put $c_i = \lfloor a_i/r \rfloor$. The inequality in (1) permits integers d_i with

$$0 \leq d_i \leq c_i, \quad \sum_i d_i = b - 1.$$

Because $r \nmid a_i$,

$$a_i - r d_i \geq a_i - r \left\lfloor \frac{a_i}{r} \right\rfloor = a_i \bmod r \geq 1. \quad (8)$$

Thus the single row $(d_i)_i$ satisfies (3)–(4), and Theorem 2 applies. $\square$

6 Sharpness and relation to width five

The allocation condition pays for every spacer increment against the worst-case remaining capacity. For $[a]_q [2]_{q^2} [2]_{q^3}$, the unique one-column allocation requires $a - 2 - 3 \geq 1$, so it activates at $a = 6$. The exact products at $a = 4$ and $a = 5$ have coefficient sequences

$$(1, 1, 2, 3, 2, 3, 2, 1, 1), \quad (1, 1, 2, 3, 3, 3, 3, 2, 1, 1),$$

respectively. Hence $a = 4$ is not unimodal, while $a = 5$ is unimodal and violates (4); a mild adaptive refinement certifies it but is omitted because it adds no width-five class. For repeated steps $[a]_q [2]_{q^2} [2]_{q^2}$, the allocation threshold is $a = 5$, while $a = 2$ already has coefficients $(1, 1, 2, 2, 1, 1)$. These examples show why no necessity or sharp-threshold conjecture is asserted.

There is nevertheless a stable bounded pattern in the one-spacer necessity regimes. For $k \leq 5$, $r \leq 6$, and all parameters at most 15, an exact census of the 33,728 rows violating (1) with $k \leq 3$

or $r \leq 3$ found a midpoint-side dip in every row. Across all 4,576 fixed $(r, a_1, \dots, a_k)$ groups, the location of the first dip was independent of b throughout the violating range, and its depth was always one or two. This is computational evidence, not an exact-boundary conjecture.

The width-five q -Fibonomial theorem requires three spacer steps 2, 3, 5 in the classes admitting a bracket decomposition. Across the 60 stable residue classes, matrix allocation reaches 18 classes, divisibility absorption alone reaches none, and the hybrid reaches the same 18. It misses 24 further decomposable classes and all 18 classes without such a decomposition. The equal counts are coincidental: the 18 reached classes and the 18 undecomposable classes are disjoint. Absorption alone cannot cover a stable class because all three spacers are nontrivial while only two ordinary factors remain; allowing absorption inside the hybrid adds no new class. Its role is nevertheless essential in Theorem 1, where it recovers the divisibility branch. The same exact class census shows that the mild adaptive refinement above does not enlarge the 18-class coverage.

Thus 24 of the 42 decomposable classes remain beyond the criterion. What additional mechanism reaches those 24 classes while retaining an aligned-center proof? A separate restricted-partition coefficient- difference argument treats all width-five classes uniformly, but does not answer that structural question. An exact characterization of multi-spacer unimodality also remains open.

7 Reproducibility and scope

The proofs use exact algebra. A standard-library checker verifies (5) by both polynomial multiplication and the independent pair partition, replays the one-spacer induction, checks the matrix induction through four spacers, tests divisibility absorption, and reproduces the sharpness and width-five coverage calculations. From the repository root run

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python3 proof/qanalog_multispacer_criterion.py
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The finite corpus is regression evidence only; it is not an assumption in either universal proof.

Theorems 1 and 2 are sufficiency results. They do not characterize all unimodal products, prove necessity beyond previously known cases, or settle general q -Fibonomial unimodality.

AI assistance. The author used AI assistance in developing the proof presentation and its verification code.

References

- [1] Brendan B. Connelly, Ezekiel Ito, Thomas C. Martinez, Olha Shevchenko, and Kacey Yang. Unimodality of q -Fibonomial coefficients for small cases. *arXiv:2605.12822v1 [math.CO]*, May 12, 2026.